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Intelligent Assistant Based on Code-Switching Detection for Improving Multilingual Communication in Algerian Discourse

From mixed, ambiguous inputs to clear, homogeneous versions

A four-layer language detection system and translation pipeline covering Darija, Modern Standard Arabic, French and English.

Abstract

Online discourse in Algeria naturally mixes Darija (Algerian Arabic), French, Modern Standard Arabic (MSA) and English, often within a single sentence and across Arabic and Latin scripts. This code-switching creates misunderstandings between generations, with the diaspora, and in professional settings, and no existing assistant robustly handles intra-sentence Algerian code-switching.

This report presents an intelligent writing assistant that detects the languages in a mixed text, rewrites it into a single selected target language, and preserves the original meaning. The system combines a custom annotated dataset, a four-layer detection pipeline (script detection, custom lexicon, fine-tuned neural models, and segmentation), and a pivot-based translation pipeline with a dedicated multi-layer solution for Darija. Fine-tuned models reach 98% accuracy on Latin-script text (RoBERTa) and 90% on Arabic-script text (DziriBERT), outperforming generic baselines. A demonstration platform exposes the pipeline with colored segment views, language distributions and a layer-by-layer “AI Expert” mode.

1. Context and Problem

Algeria’s online discourse naturally mixes Darija, French, Standard Arabic and English. This phenomenon, known as code-switching, is part of everyday communication but creates real friction:

- Misunderstandings arise between generations, with the diaspora, and in professional settings.
- Darija has multiple writing styles (Arabic or Latin script, informal spellings), which makes processing harder.
- **Gap:** no existing assistant robustly handles intra-sentence Algerian code-switching.

2. Objectives and Target Users

The goal is to move from mixed, ambiguous inputs to clear, homogeneous versions. The project builds an intelligent writing assistant that:

- detects the languages in mixed text;
- rewrites the text into a single selected target language (Standard Arabic, French, English or Darija);
- preserves the original meaning.

Four groups of users are targeted:

User group	Use case
Individuals	Daily messaging
Professionals	Formal text
Diaspora	Staying connected
Language learners	Educational use

3. What Does Code-Switching Look Like?

The following input illustrates intra-sentence code-switching, mixing Darija and French in a single sentence:

> **Original input:** *“Rah nroh l’université demain, j’ai un examen important, inshallah njib bonne note”*

Detection breakdown

Segment	Detected language
Rah nroh	Darija (Arabic script form: راه نروح)
l’université …	French
inshallah	Darija / Arabic
njib …	Darija + French mix

The four homogeneous versions

Standard Arabic (MSA)

لدي امتحان مهم، بالتوفيق سأحصل على علامة جيدة سأذهب إلى الجامعة غدا،

French (FR)

Je vais à l’université demain, j’ai un examen important, j’espère avoir une bonne note

English (EN)

I’m going to university tomorrow, I have an important exam, hopefully I’ll get a good grade

Darija (DZA)

غدوة، عندي امتحان مهم، إن شاء نجيب نوتة زوينة ران نروح للجامعة

4. Related Work and Why Existing Solutions Fall Short

Existing resources address parts of the problem but not Algerian intra-sentence mixing. The observed outcome was poor performance on Algerian intra-sentence code-switching.

Dataset	Languages	Script	Intra-sentence	Notes
CAFE	FR, AR	Mixed	Limited	Focus on formal text
ELNER-DZ	Darija, FR	Latin	Low	Entity focus, poor mixing
Algerian-Darija	Darija	Arabic	None	Monolingual dialect focus

Key reasons

- Algerian code-switching is highly context-specific.
- Darija has multiple writing styles (Arabic/Latin, informal spellings).
- Borrowings and ambiguous words are not handled.
- Coverage of true intra-sentence mixing is limited.

5. Custom Dataset

Because existing datasets gave poor results, a custom dataset was built using a two-pronged approach: (1) manual annotation of 2,550 sentences, and (2) automatic extraction followed by manual verification.

Dataset characteristics

- Multilingual, mixed-script, real Algerian discourse.
- Word- and segment-level language labels.
- Script information.
- Entity tagging, used for translation robustness.

Data pipeline

Source collection	→	**Cleaning**	→	**Annotation**	→	**Verification**	→	**Train / Dev / Test split**

6. Four-Layer Language Detection System

The input text passes through four successive layers, producing labeled segments and language statistics.

Layer	Component	Role
1	**Script detection**	Arabic vs Latin character filtering and classification (initial script filtering)
2	**Lexicon**	Custom dictionary lookup for Darija, French, English and MSA (dictionary verification)
3	**AI models**	Fine-tuned DziriBERT and RoBERTa models, one per script (neural classification)
4	**Segmentation**	Ambiguity resolution and coherent grouping (contextual smoothing)

6.1 Layer 1: Script detection

- Rule-based detection of Arabic vs Latin characters.
- Handles mixed-script sentences quickly.

6.2 Layer 2: Custom lexicon

- Covers Darija (Arabic and Latin variants), French, English and MSA.
- Supports borrowings and context-independent identification.
- Resolves ambiguity, for example the word ‘table’ shared by French and English.

Example lexicon entries

Entry	Language tags	Variants	Script
Table	FR, EN	table, tables	Latin
Nroh	Darija	nroh, nrouh, نروح	Mixed (Latin/Arabic)
Inshallah	Darija, MSA	inshallah, nchallah, إن شاء □	Mixed

7. Models and Results

Layer 3 relies on two fine-tuned models, one for each script family:

- **DziriBERT** (Arabic script): specialized for MSA and Darija in Arabic script, reaching **90%** accuracy.
- **RoBERTa** (Latin script): multilingual support for Darija, French and English, reaching **98%** accuracy.

NER add-ons (*marefa-nlp/marefa-ner* and *hatmimoha/arabic-ner*) preserve named entities during rewriting and translation. A Google Gemini API fallback exists but is rarely triggered.

Model comparison

Model type	Scope	Outcome
langdetect / langid	General	**Rejected** (poor code-switching performance)
MARBERT alone	Arabic	**Insufficient** (Darija)
DziriBERT (fine-tuned)	DZA_AR + MSA	**90% accuracy**
RoBERTa (fine-tuned)	DZA_LAT + FR + EN	**98% accuracy**

Baselines were rejected due to poor performance on intra-sentence code-switching.

Detection performance

Model	Script	Languages	Accuracy
DziriBERT	Arabic	MSA, Darija	**90%**
RoBERTa	Latin	Darija, FR, EN	**98%**

Key takeaways: The Gemini fallback is almost never triggered. Custom training combined with the lexicon improves robustness, and the system outperforms generic baselines across all Algerian discourse scripts.

8. Segmentation and Ambiguity Resolution

Layer 4 groups token-level labels into coherent language segments, which enables cleaner rewriting and translation by chunk. Ambiguity is handled through dictionary cross-checks, continuity heuristics, and local context analysis of the previous and next tokens.

FRENCH	ENGLISH
J'ai une table	**and two chairs**

Ambiguity resolution: the word *"table"* exists in both French and English. Contextual heuristics identify it as French because of the preceding tokens *"J'ai une"*.

9. Generating Homogeneous Versions

Translation follows a pivot approach: all detected segments are protected for entities, translated to English (the pivot), and then generated into the four target languages.

Detected segments	→	Entity protection	→	Pivot: English	→	Target generation (4 outputs)

Other targets (FR / EN / MSA)

Fine-tuned NLLB and MarianMT provide high-quality, well-supported output generation.

Darija translation challenge

Darija suffers from low resources and the absence of standardized translation models. A multi-layer solution is used:

- custom dictionary lookup for frequent words;
- Gemma 2 for approximate translation;
- refinement with fine-tuned NLLB / MarianMT.

10. Demonstration Platform

Input methods

- Manual text input.
- File upload (TXT, PDF, DOC).
- Voice input (Whisper).

Key features

- Layer-by-layer explanation (“AI Expert” mode).
- Colored segment view for code-switching visualization.
- Language percentage distribution and per-token details.
- Generation of 4 homogeneous versions.

Example distribution (platform output)

DZA_AR	MSA	DZA_LAT	FR	EN	ENTITY
5%	0%	24%	43%	19%	10%

11. Complete Example Walkthrough

> **Input:** *“Rah nroh l’université demain, j’ai un examen important, inshallah njib bonne note”*

Detection results: Darija (DZA) 29%, French (FR) 53%, English (EN) 18%.



The four generated outputs are the same as those shown in Section 3.

12. Difficulties and Solutions

Difficulty	Solution
Poor results with existing datasets	Created a custom annotated dataset

Difficulty	Solution
Darija not recognized by standard models	Fine-tuned DziriBERT and RoBERTa
Ambiguous words	Dictionary and context analysis
Darija translation resources lacking	Dictionary → Gemma 2 → NLLB / MarianMT
Model explainability	Layer-by-layer display

> **Engineering around low-resource constraints is the core novelty of this work.**

13. Conclusion

Key achievements:

- Reliable detection: 98% on Latin script and 90% on Arabic script.
- Lexicon combined with fine-tuning outperforms generic baselines.
- Handles intra-sentence code-switching and mixed scripts.
- Explainable interface through a layer-by-layer display.
- Voice support via Whisper.

The work covers dataset design and manual annotation with evaluation, model fine-tuning (DziriBERT and RoBERTa) with NER integration, and the translation pipeline with the demonstration application and user interface.

14. Future Directions

- Improve Darija translation quality and its evaluation.
- Expand the dataset (sources and domains).
- Provide an API for third-party integration.
- Build a mobile application.
- Add languages (Spanish, Turkish).
- Add real-time messaging integration with a user feedback loop.

Points for further discussion

- Why not simply use a general-purpose large language model such as GPT-4?
- How should out-of-dictionary words be handled?
- How is Darija translation evaluated?
- What are the limitations of the system?